C – SYLLABUS

1. C – Introduction
2. History
3. Fundamentals
4. Programming structure
5. Printf, variables, scanf
6. Control structures / control statements If, if..else, if…else if ladder, nested if, switch, while, do..while, for, goto label
7. Arrays
8. strings
9. Pointers
10. Dynamic memory allocation
11. User defined functions
12. Recursion
13. Storage classes
14. Preprocessor
15. Structure
16. Union
17. Enum
18. Typedef
19. File handling with project work
20. Command line arguments

Fee Details:

Rs : 2,000/- INR (Without videos means you will get Live sections & Class Notes Only)

Rs : 4,000/- INR (With videos means you will get Live sections, Recorded Videos & Class Notes)

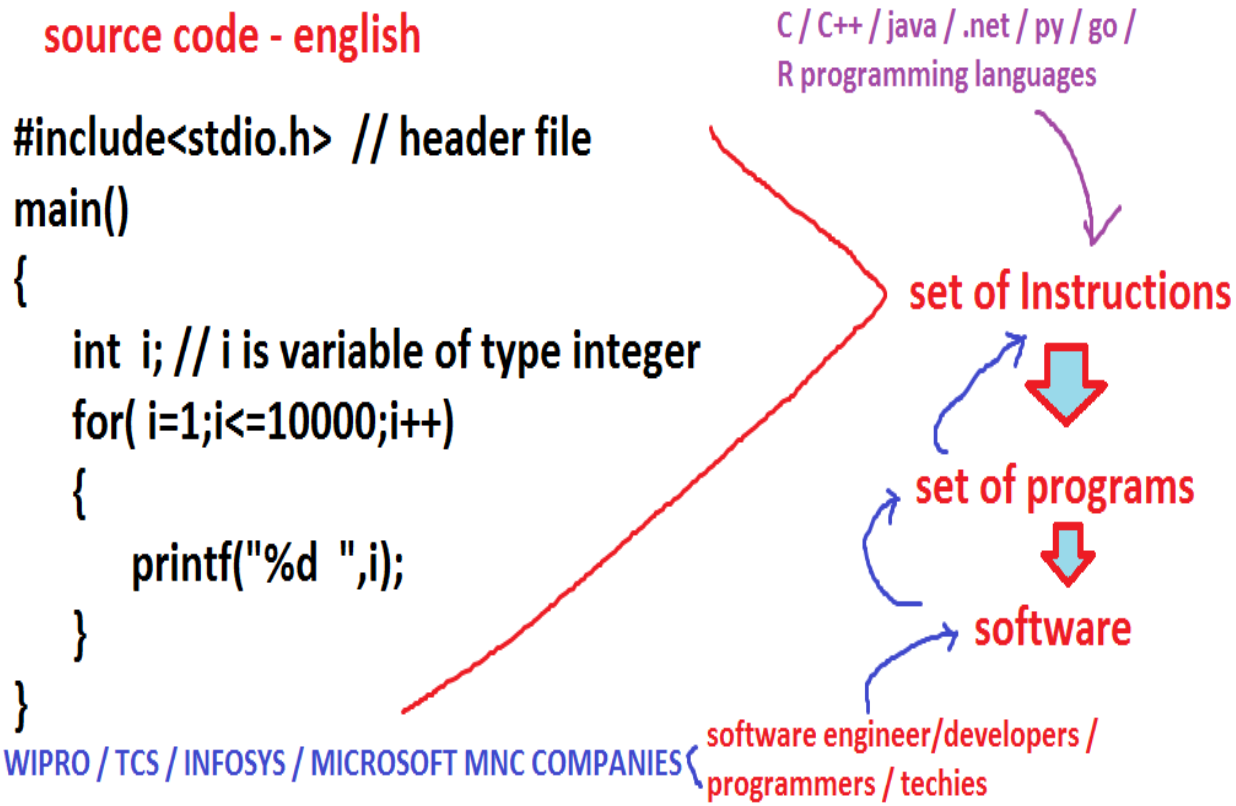
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INTRODUCTION TO C

C is a

1. **It is a high level / middle level programming language.**
2. **C is a compiler based programming language.**
3. **C is a general purpose or multipurpose programming language.**
4. **C is a procedure oriented programming [pop] language.**



What is a program?

Set of instructions are called program.

What is a software?

Set of program is called software.

Again the software is divided into 2 types.

1. **System software**
   Eg: operating system, device drivers, translators
2. **Application software.**
   Eg: whatsapp, fb, instagram,googlepay,….

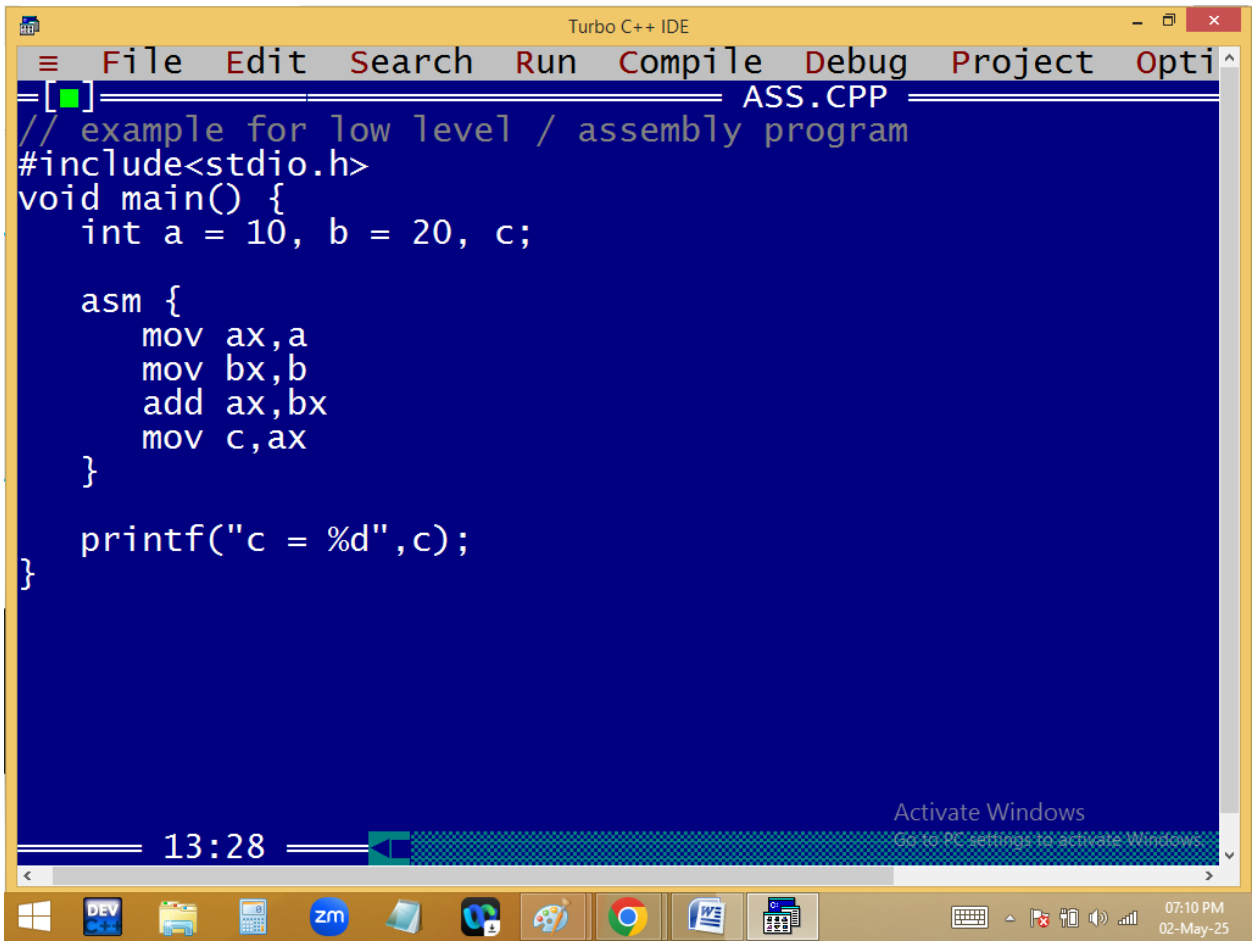
What is a language?

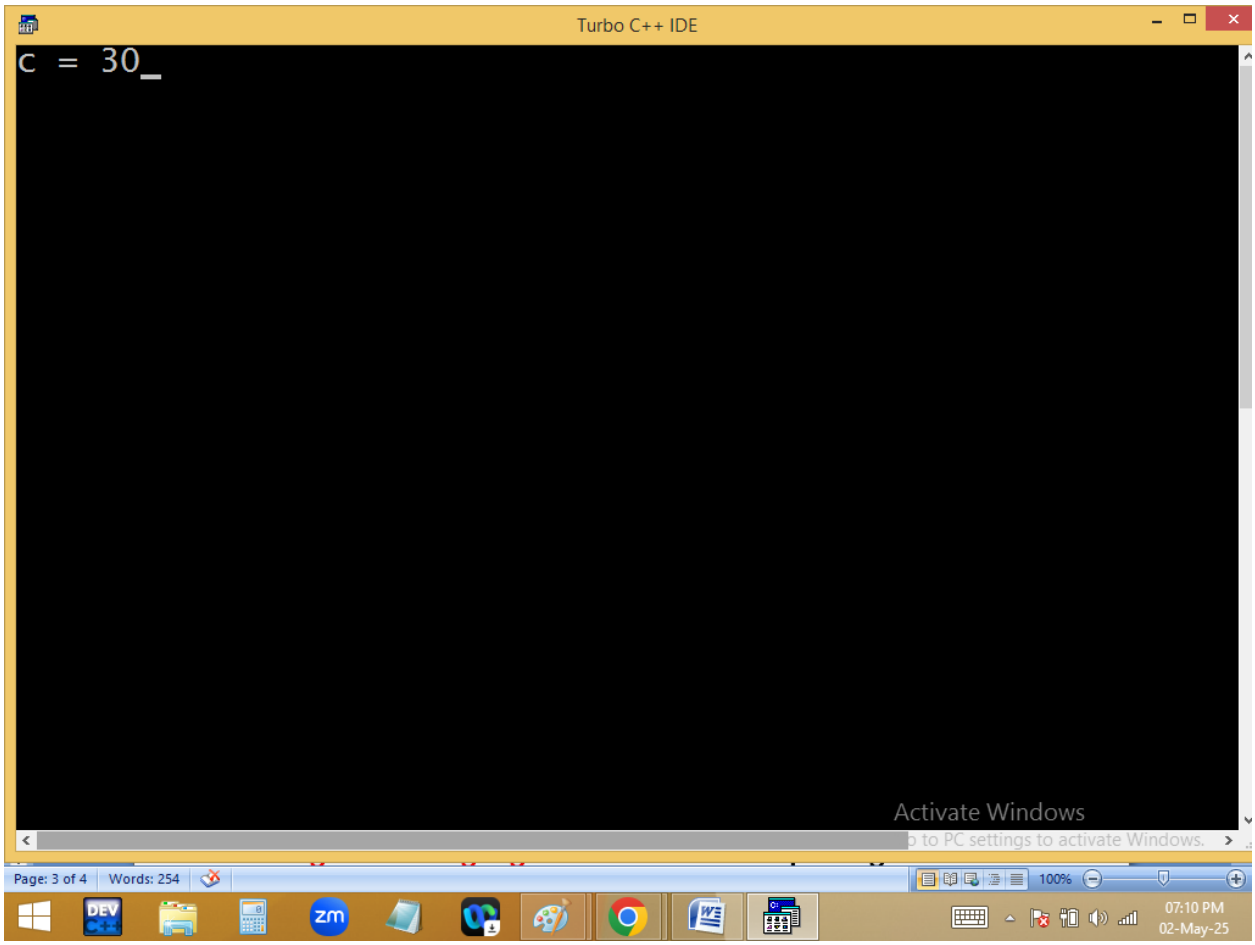
Generally the languages are used to communicate with others. For example the languages like telugu / English / hindi / Marathi etc are called human languages and which are used to communicate with the humans.

To communicate with the machines / computers are using the programming languages like C / C++ / java / .net / py etc. by using these languages we can create the software [programs] to communicate with the machines , which are doing our work in easy , efficiency way.

Basically the languages divided into 3 types.

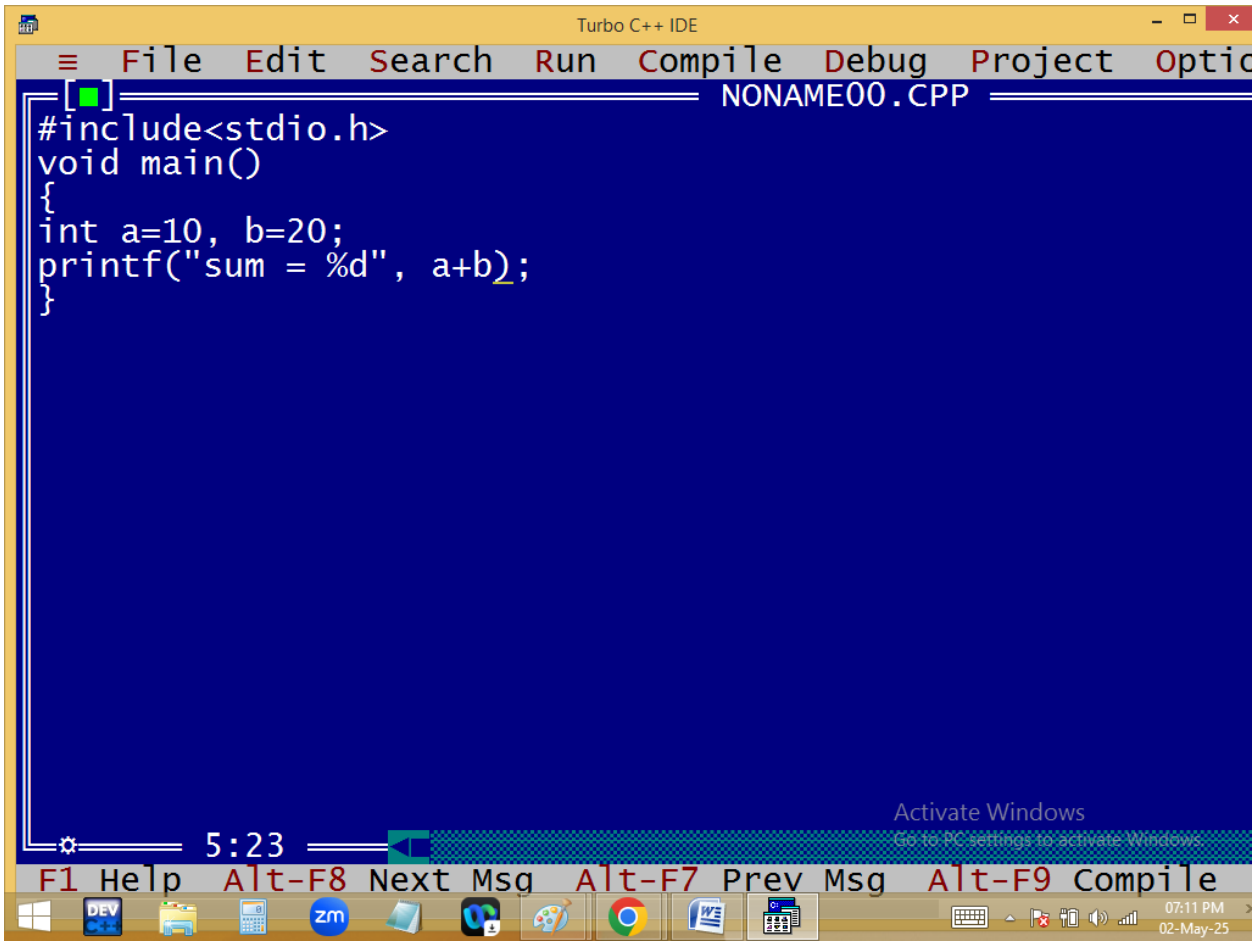
1. **Machine language / binary code: Created with binary code and very difficult for humans.**
   Eg: 10001111
2. **Low level / assembly language: Created with English like shortcuts called MNEMONICS.**
   Eg: gd evng, sub, add,..

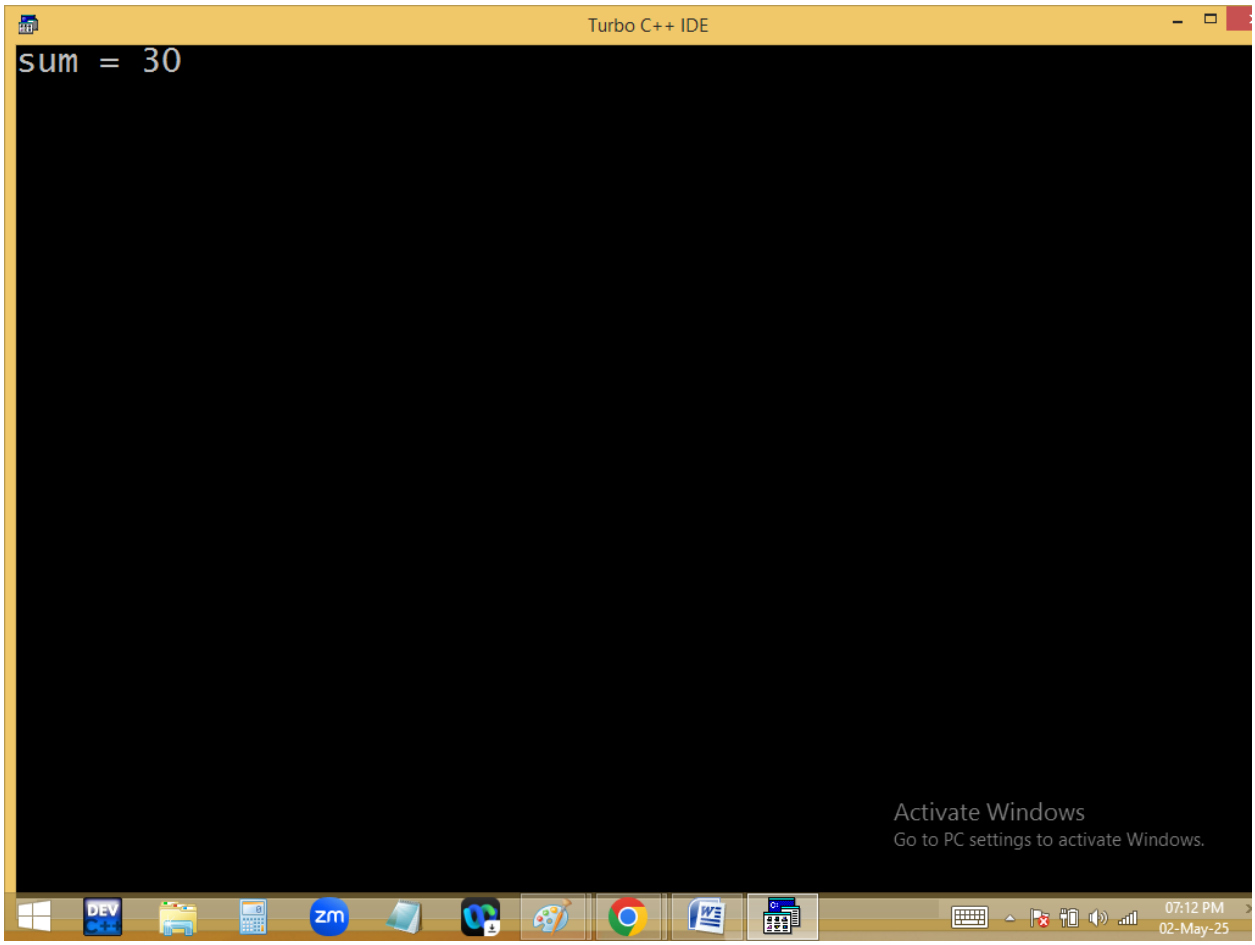

Turbo C++ IDE
File Edit Search Run Compile Debug Project Opti
ASS.CPP
// example for low level / assembly program
#include<stdio.h>
void main() {
int a = 10, b = 20, c;
asm {
mov ax,a
mov bx,b
add ax,bx
mov c,ax
}
printf("c = %d",c);
}
13:28
Activate Windows
Go to PC settings to activate Windows.
07:10 PM
02-May-25



3. **High level language:** Created with simple English and easy to understand.

   Eg: good evening, subject, address,….


Turbo C++ IDE
File Edit Search Run Compile Debug Project Optic
NONAME00.CPP
#include<stdio.h>
void main()
{
int a=10, b=20;
printf("sum = %d", a+b);
}
5:23
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Go to PC settings to activate Windows.
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile
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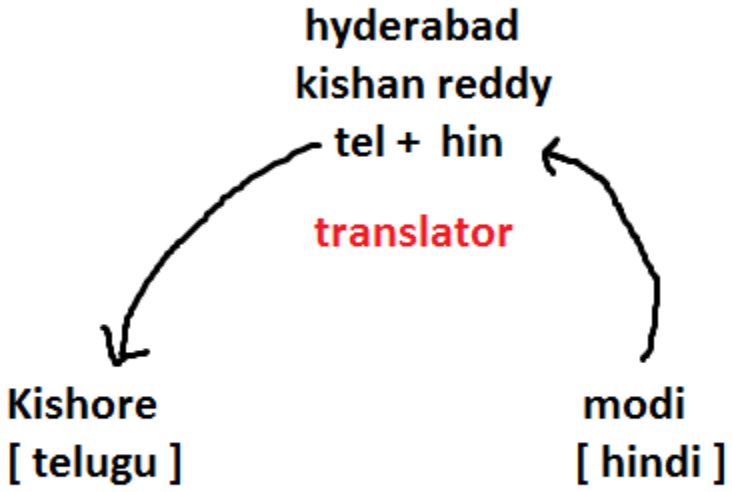
C is basically high level language with low level features. Hence it is called middle level language also.

C low level features are used to design system software and high level features used to design application software. Hence c is a multi-purpose programming language.

Naveen – 7337313415

5-5 notes:

What is translator?



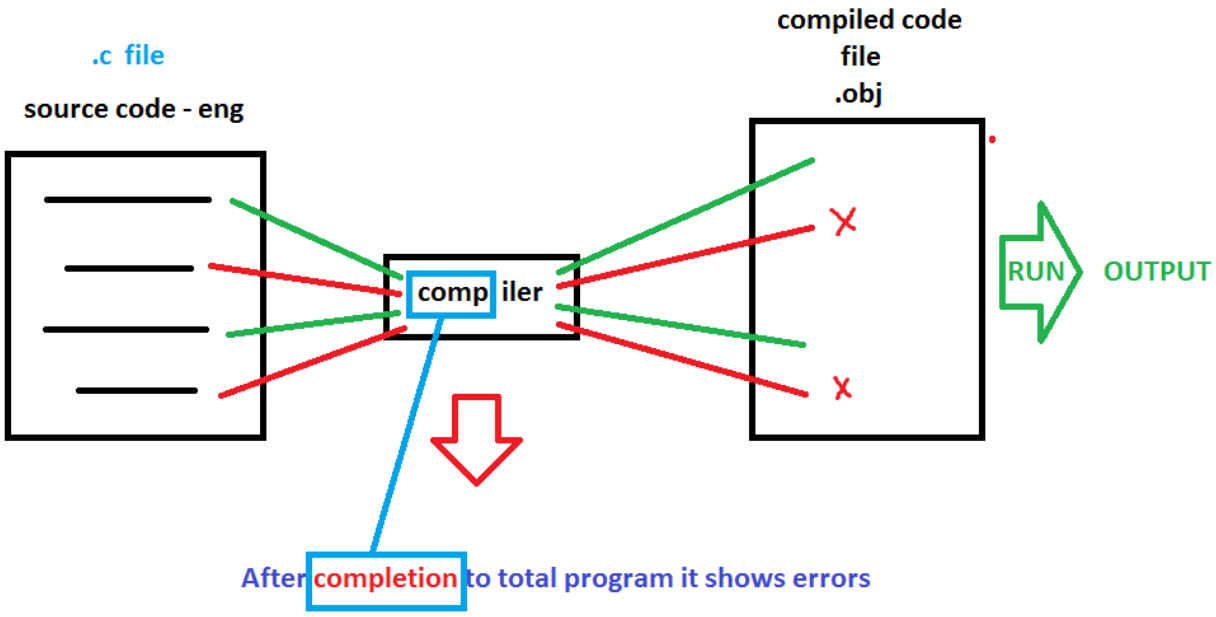
Always the programmer given instructions are in English, which is called source code or source program. But the computer understandable code is binary code.

To convert this source into binary code and to check the programming mistakes [errors] we are using 3 translators.

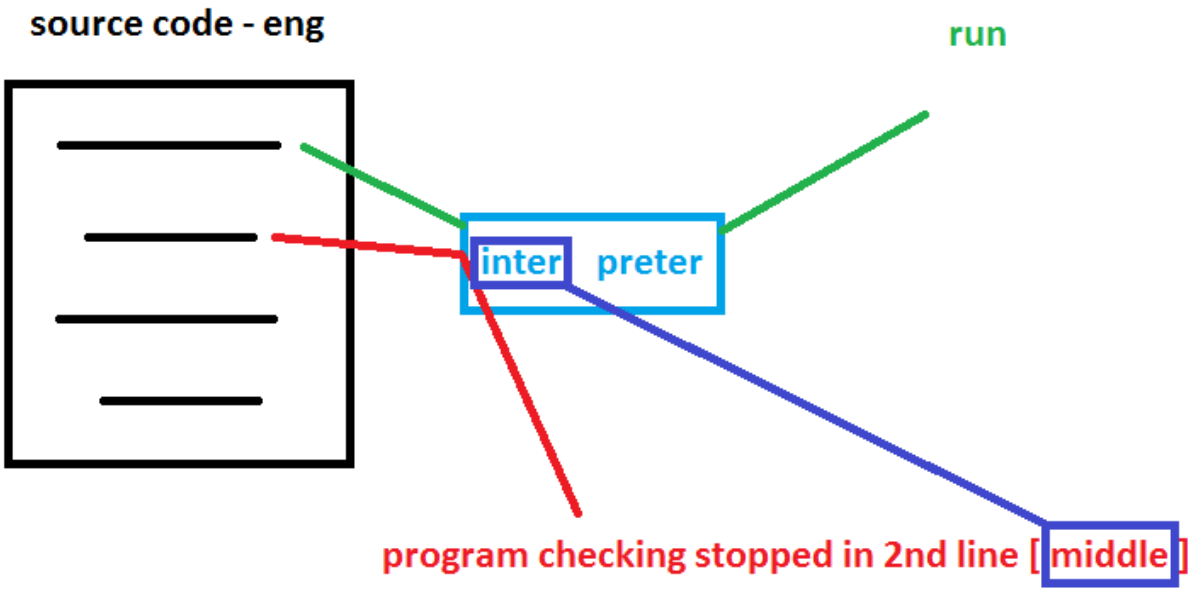
1. **Compiler**
2. **Interpreter**
3. **Assembler**

Compiler and interpreter used to convert high level programs to binary code.

Compiler completes the entire source code into binary code at once by leaving error lines.



Interpreter checks line by line.



Assembler is used to convert low level / assembly language to binary code.

Assembler working style is similar to the compiler.

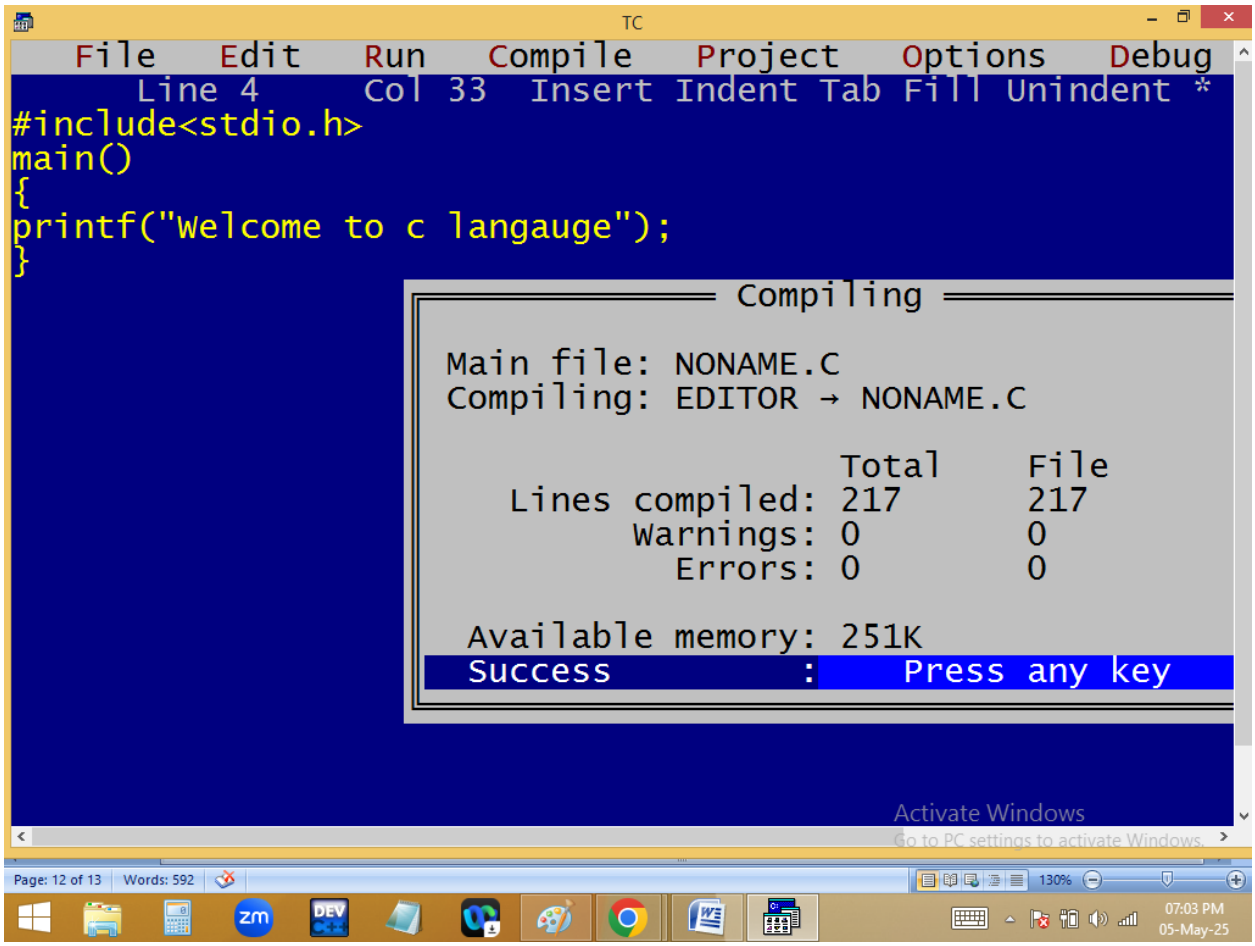
In c & C++ we are using compiler. Hence they are called compiler based programming languages.

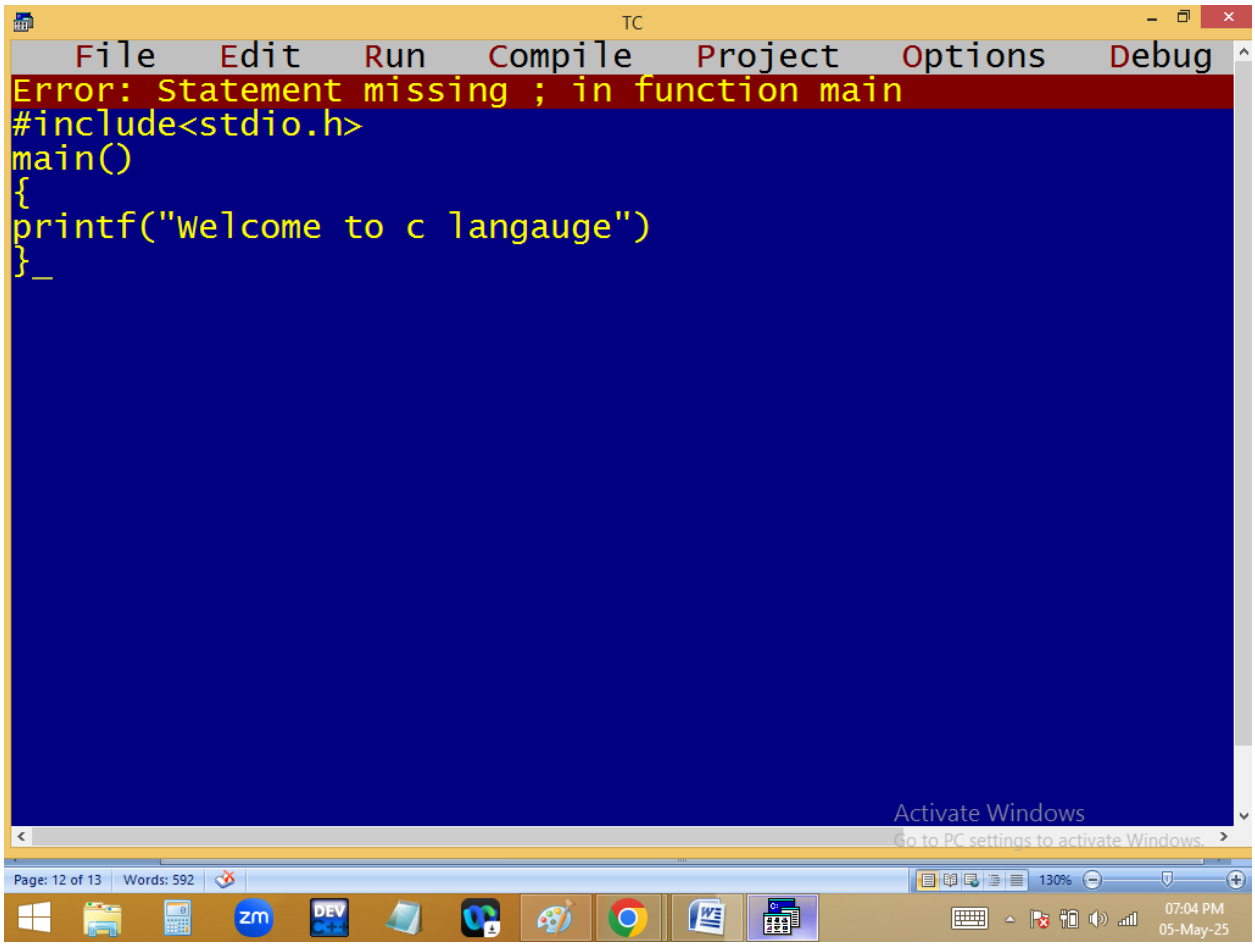
But in java we are using compiler and interpreter both. Hence it is a compiler based interpreted language.

Basically the errors divided into 2 types.

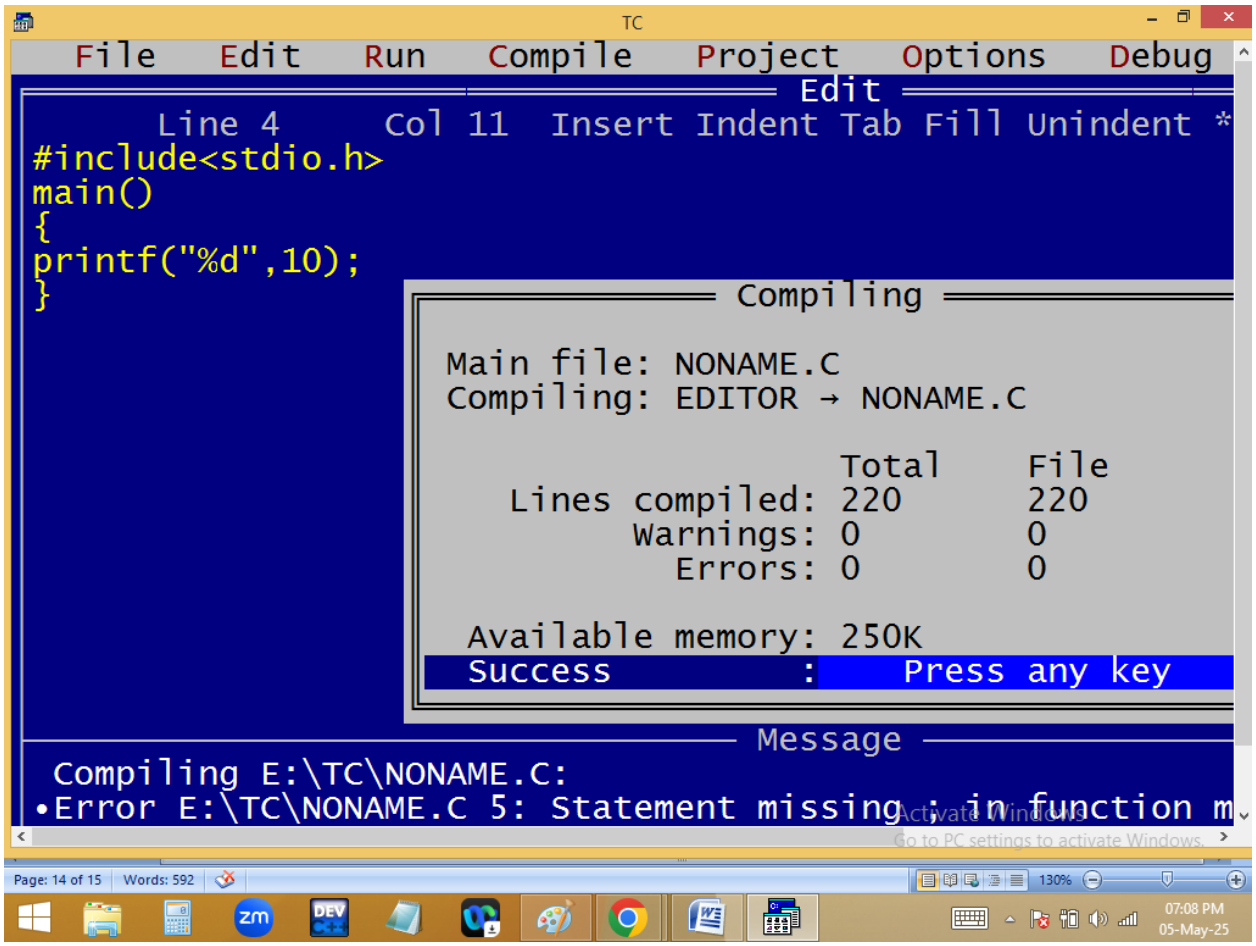
1. **Syntax error / compile errors: When the programmer is not following the language rules then we are getting compile time errors**

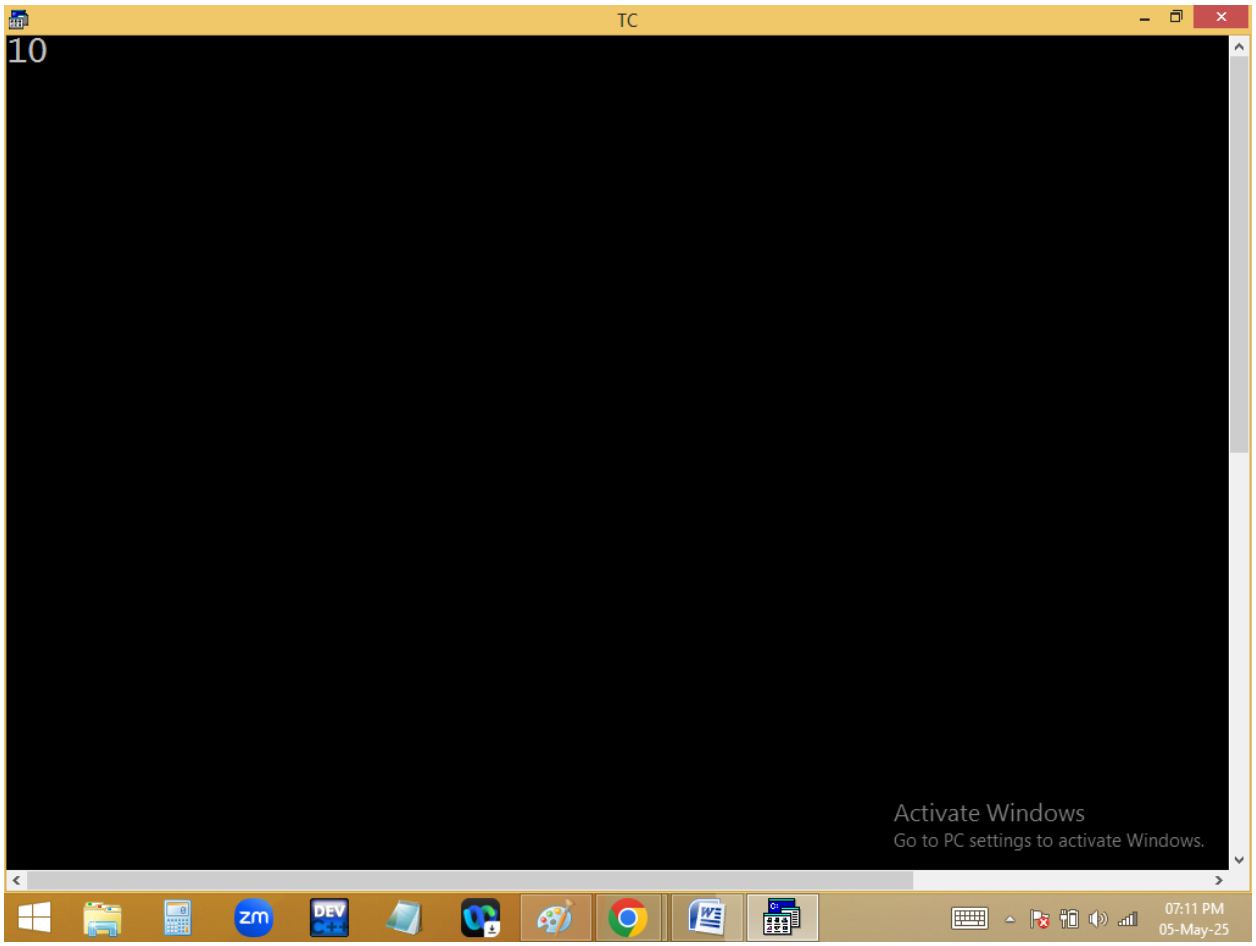
   Example:

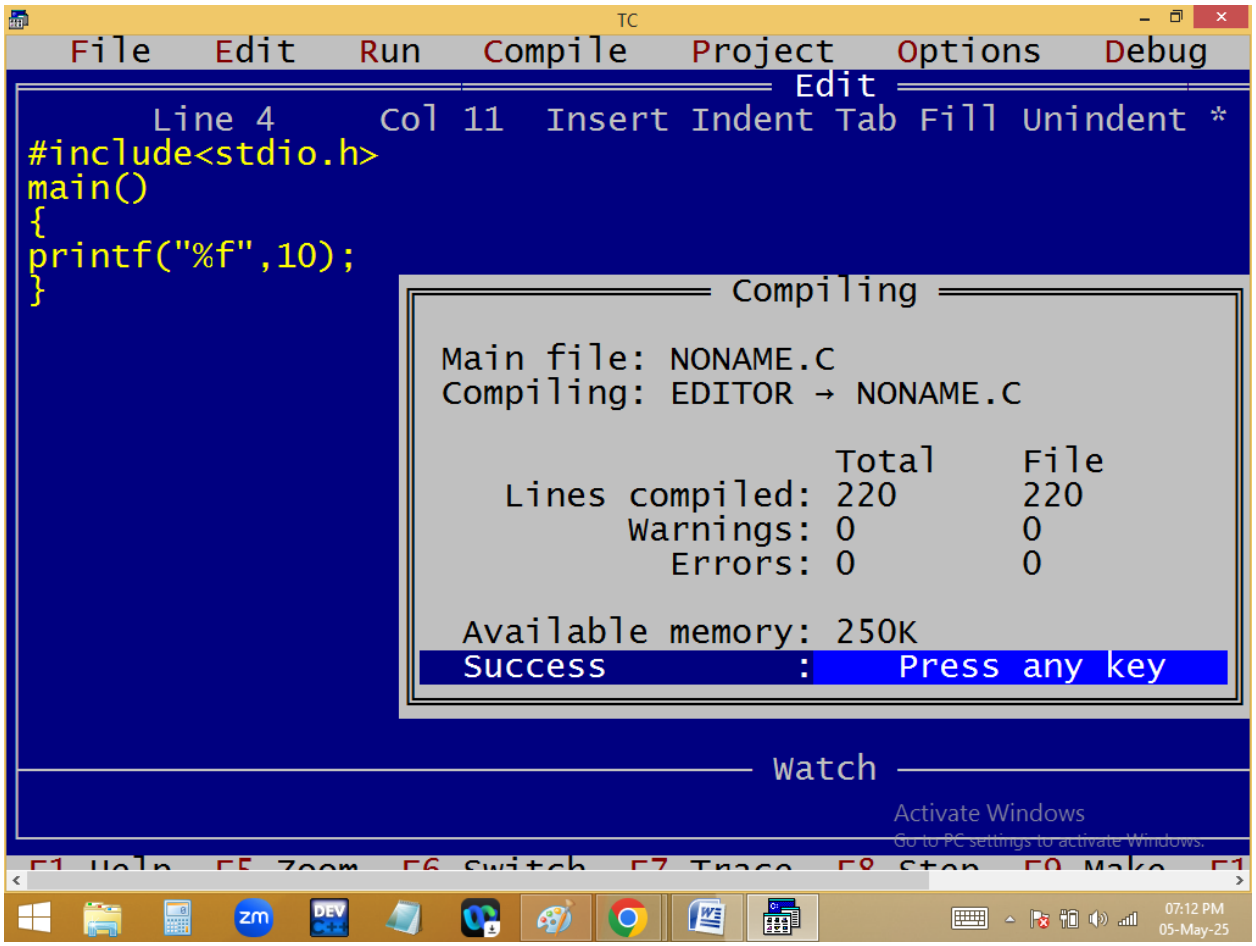

TC
File Edit Run Compile Project Options Debug
Line 4 Col 33 Insert Indent Tab Fill Unindent *
#include<stdio.h>
main()
{
printf("Welcome to c langauge");
}
Compiling
Main file: NONAME.C
Compiling: EDITOR → NONAME.C
Total File
Lines compiled: 217 217
Warnings: 0 0
Errors: 0 0
Available memory: 251K
Success : Press any key
Activate Windows
Go to PC settings to activate Windows.
Page: 12 of 13
Words: 592
130%
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05-May-25

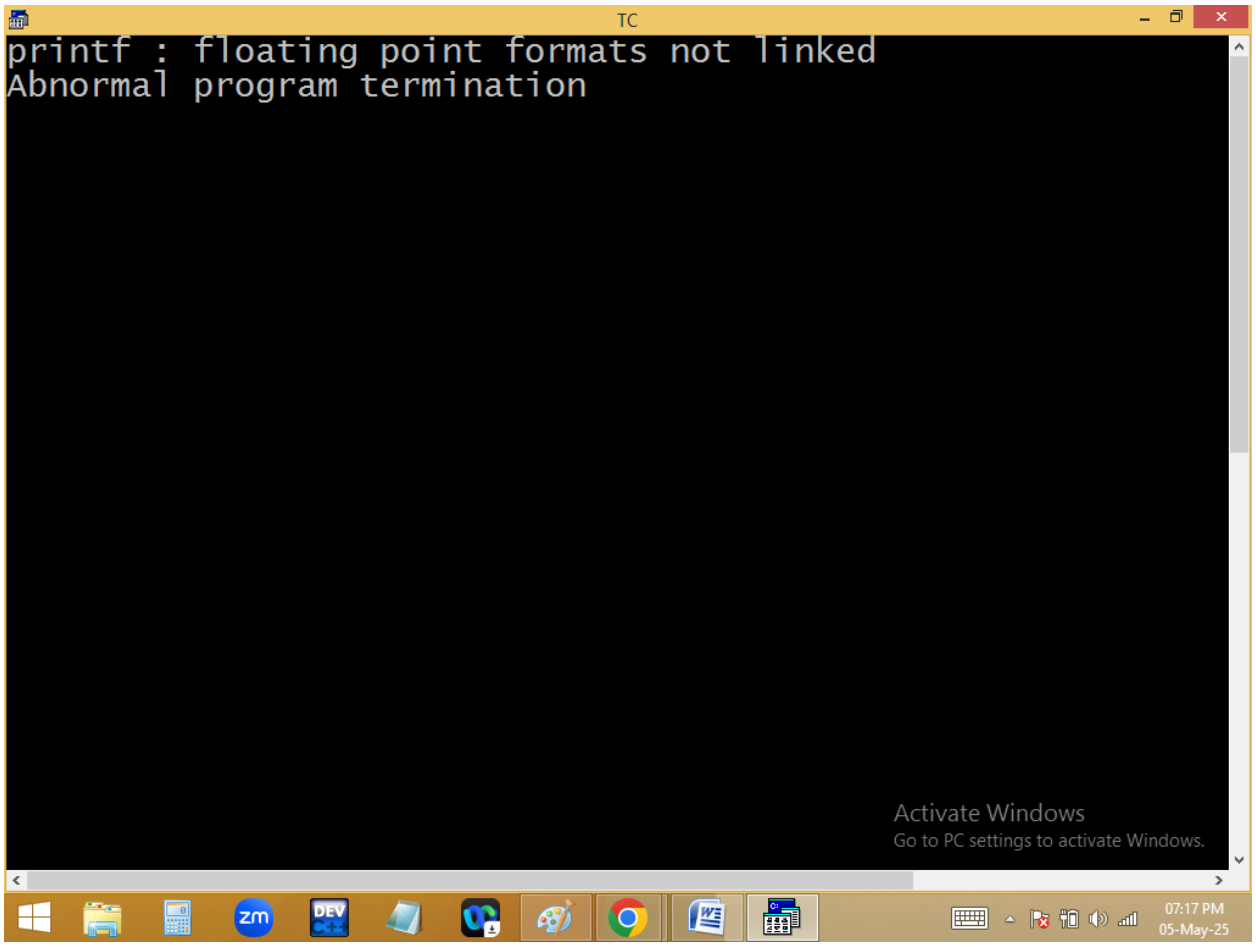


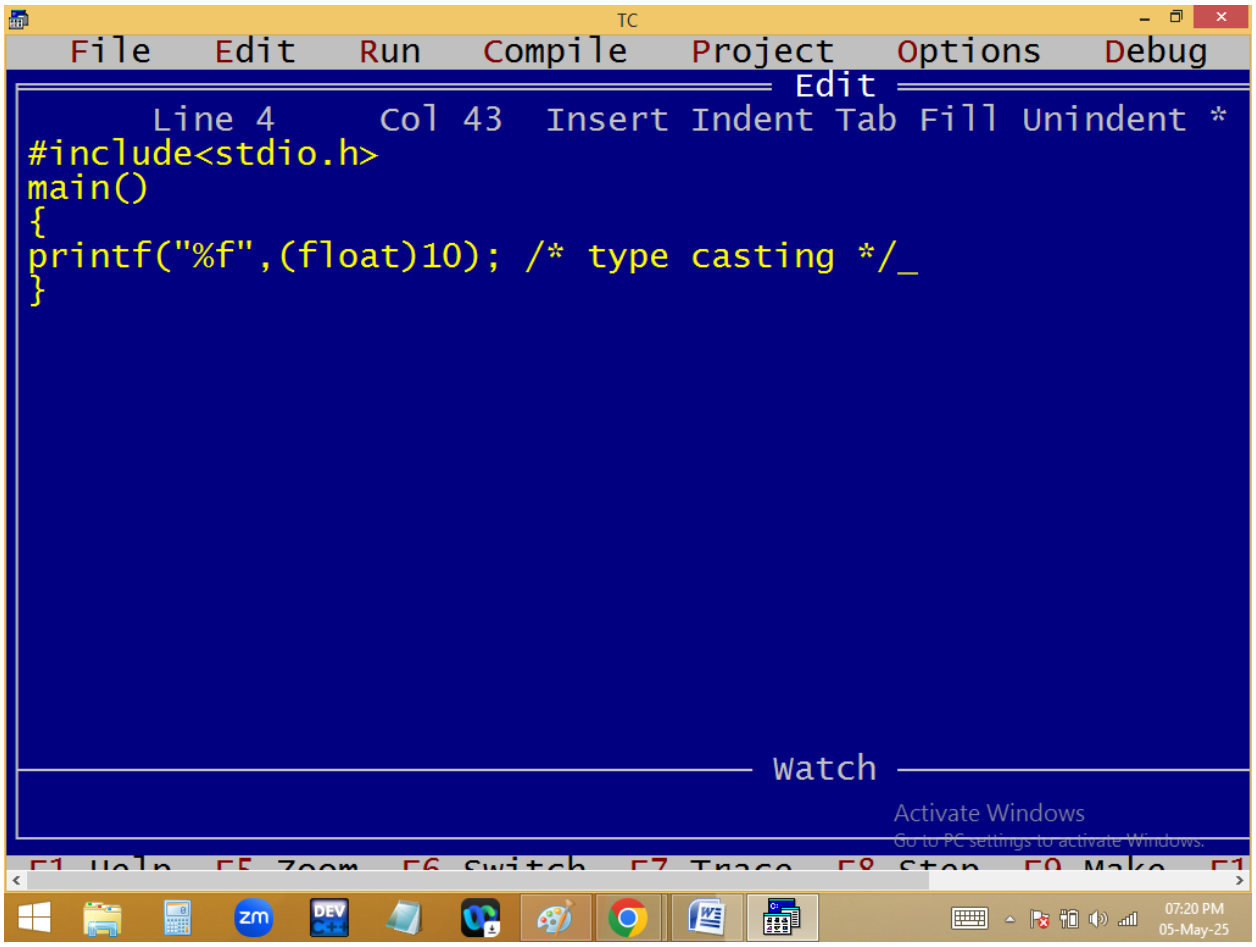
2. **Runtime errors: Programmer is following all the rules but the system notable deliver the answer, then it throws runtime errors.**

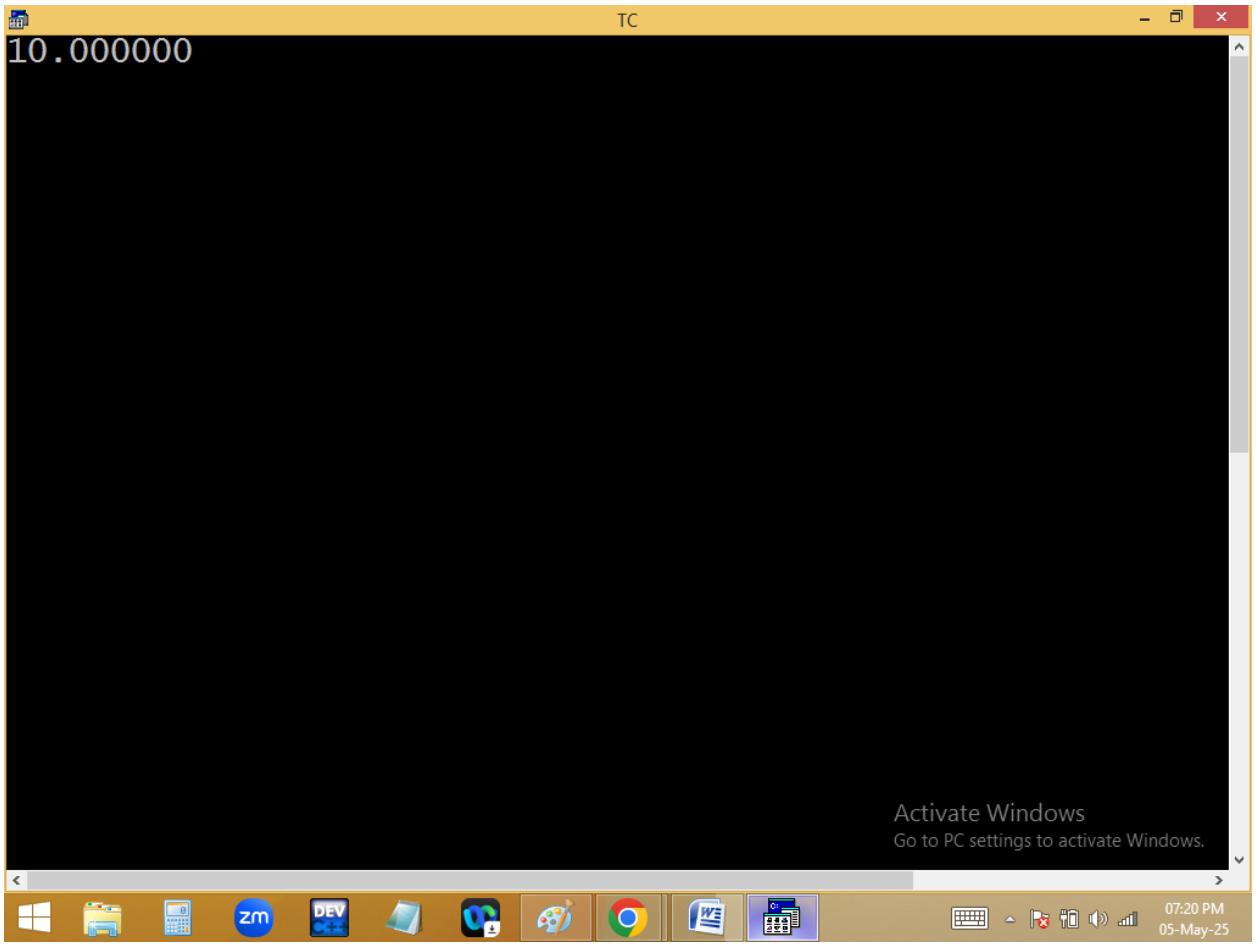

TC
File Edit Run Compile Project Options Debug
Edit
Line 4 Col 11 Insert Indent Tab Fill Unindent *
#include<stdio.h>
main()
{
printf("%d",10);
}
Compiling
Main file: NONAME.C
Compiling: EDITOR → NONAME.C
Total File
Lines compiled: 220 220
Warnings: 0 0
Errors: 0 0
Available memory: 250K
Success : Press any key
Message
Compiling E:\TC\NONAME.C:
•Error E:\TC\NONAME.C 5: Statement missing ; in function m
Activate Windows
Go to PC settings to activate Windows.
Page: 14 of 15
Words: 592
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TC
10
Activate Windows
Go to PC settings to activate Windows.
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05-May-25

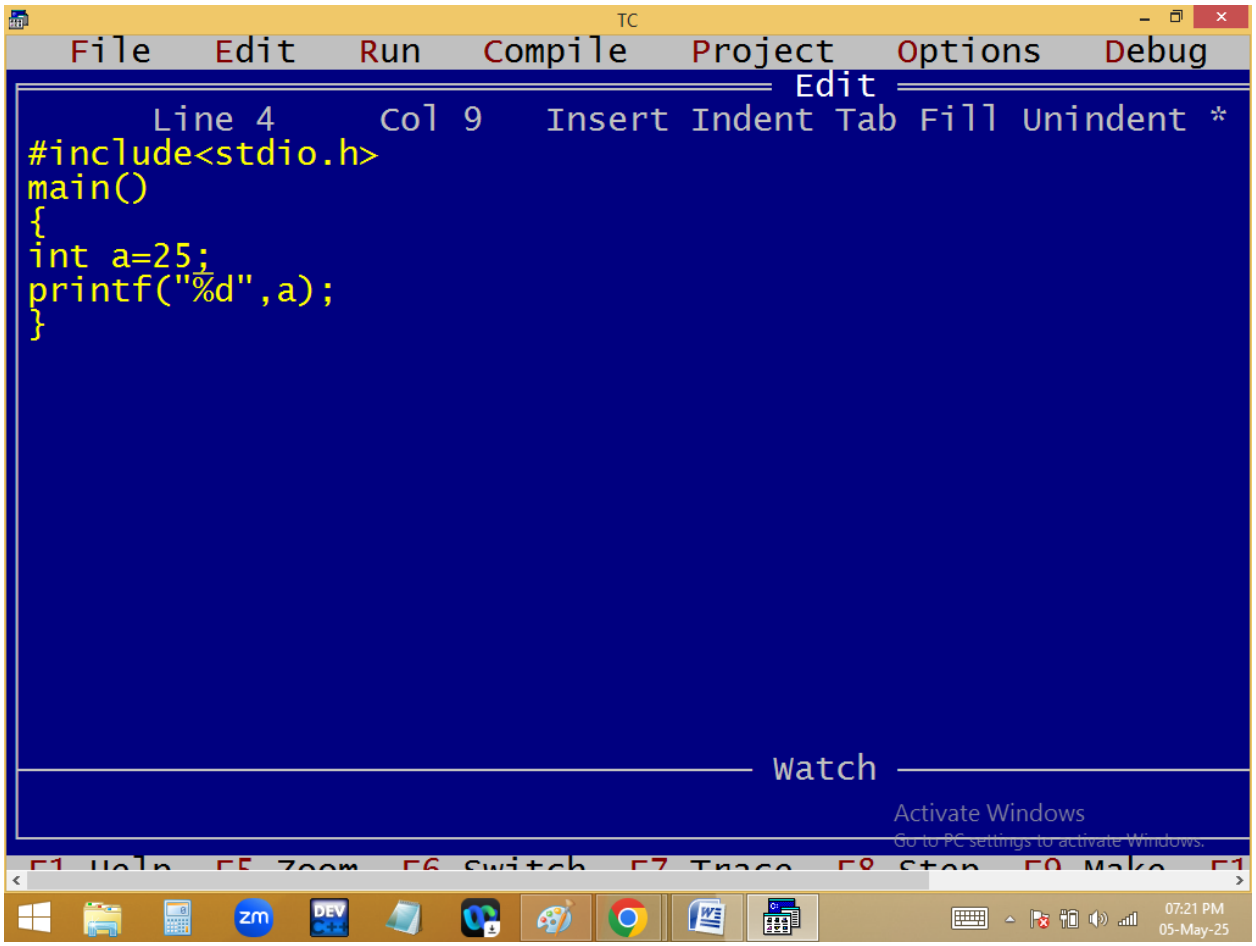

TC
File Edit Run Compile Project Options Debug
Edit
Line 4 Col 11 Insert Indent Tab Fill Unindent *
#include<stdio.h>
main()
{
printf("%f",10);
}
Compiling
Main file: NONAME.C
Compiling: EDITOR → NONAME.C
Total File
Lines compiled: 220 220
Warnings: 0 0
Errors: 0 0
Available memory: 250K
Success : Press any key
Watch
Activate Windows
Go to PC settings to activate Windows.
07:12 PM
05-May-25

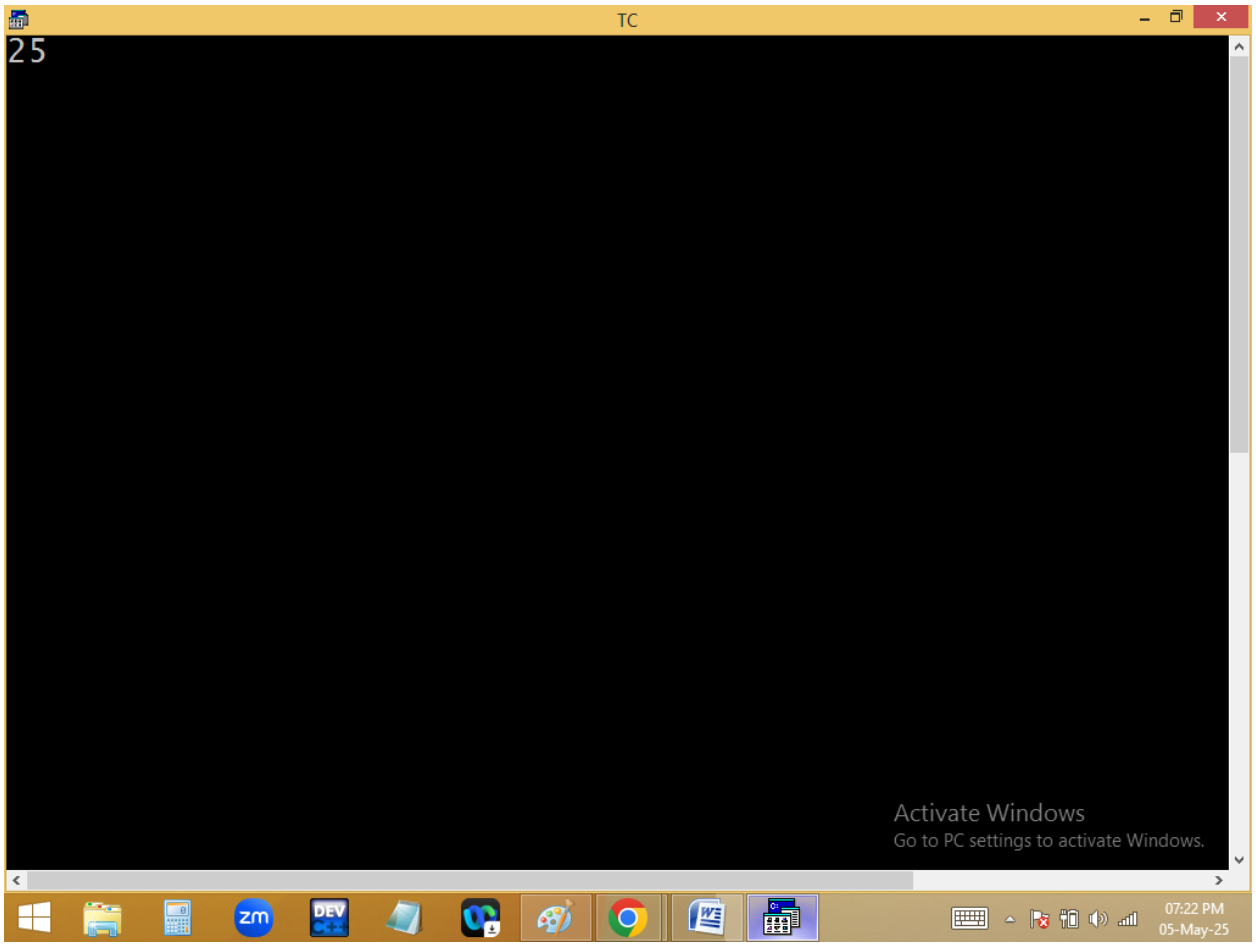

TC
printf : floating point formats not linked
Abnormal program termination
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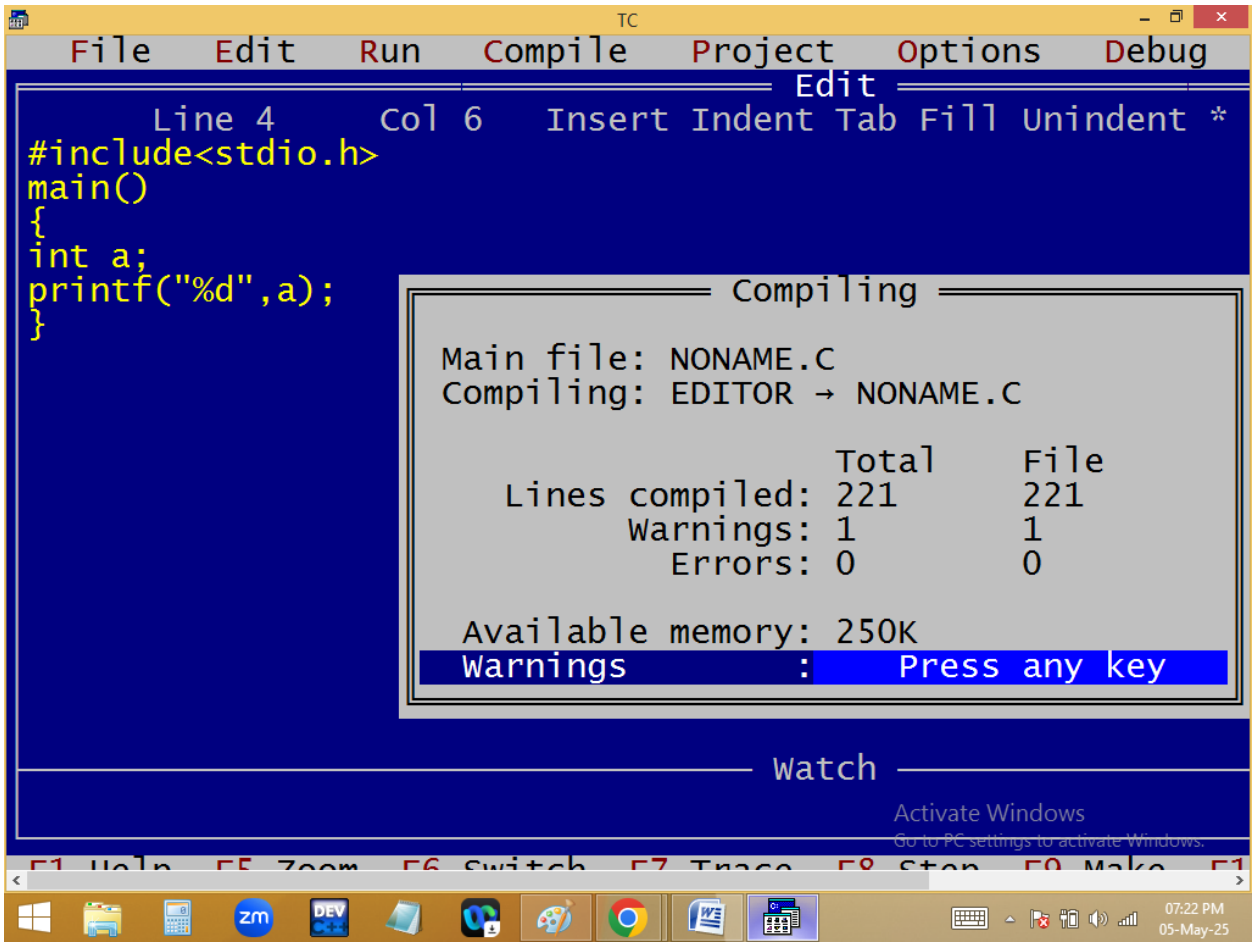

TC
File Edit Run Compile Project Options Debug
Edit
Line 4 Col 43 Insert Indent Tab Fill Unindent *
#include<stdio.h>
main()
{
printf("%f",(float)10); /* type casting */_
}
Watch
Activate Windows
Go to PC settings to activate Windows.
07:20 PM
05-May-25

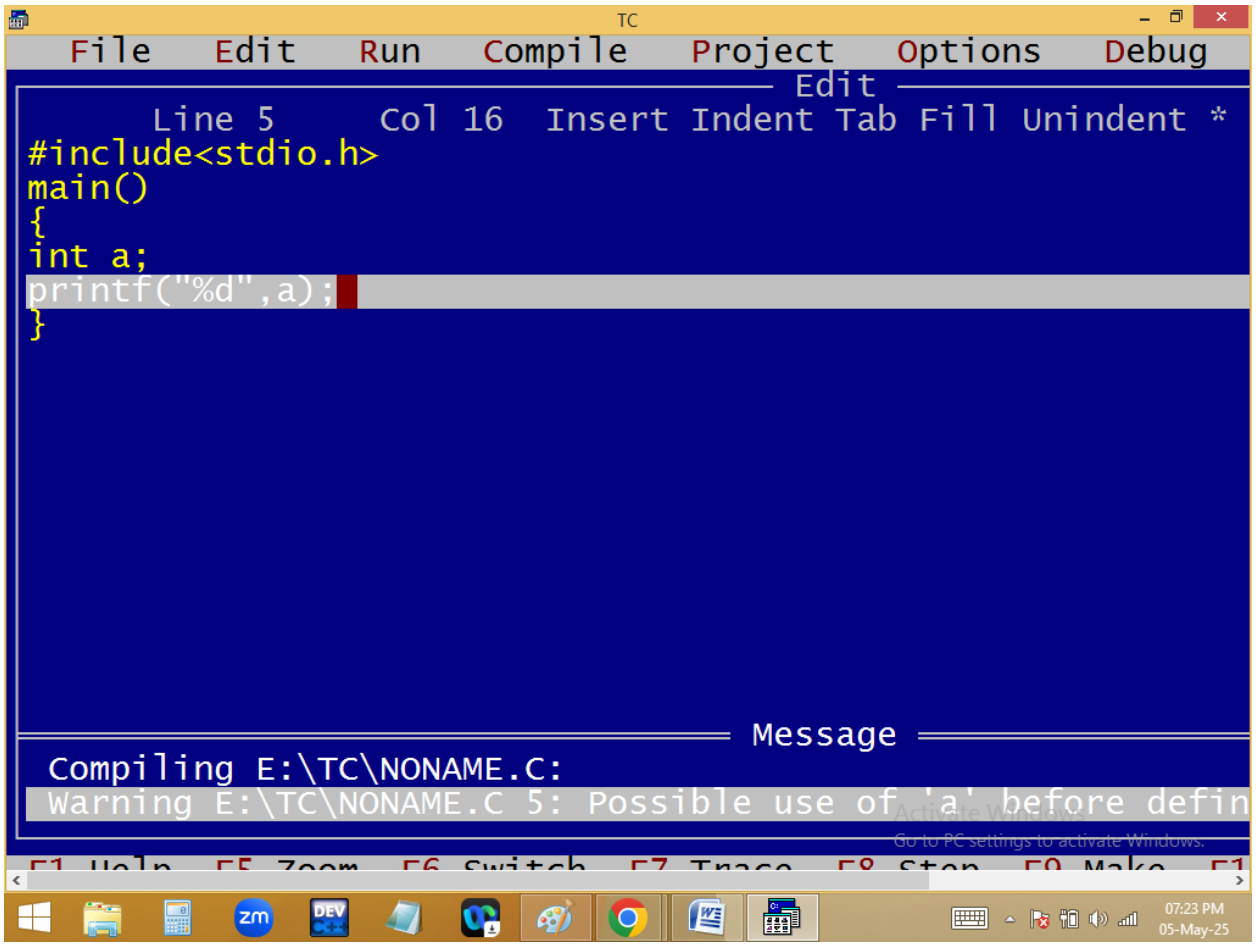


Example for warning?


TC
File Edit Run Compile Project Options Debug
Edit
Line 4 Col 9 Insert Indent Tab Fill Unindent *
#include<stdio.h>
main()
{
int a=25;
printf("%d",a);
}
Watch
Activate Windows
Go to PC settings to activate Windows.
07:21 PM
05-May-25


TC
25
Activate Windows
Go to PC settings to activate Windows.
07:22 PM
05-May-25


TC
File Edit Run Compile Project Options Debug
Edit
Line 4 Col 6 Insert Indent Tab Fill Unindent *
#include<stdio.h>
main()
{
int a;
printf("%d",a);
}
Compiling
Main file: NONAME.C
Compiling: EDITOR → NONAME.C
Total File
Lines compiled: 221 221
Warnings: 1 1
Errors: 0 0
Available memory: 250K
Warnings : Press any key
Watch
Activate Windows
Go to PC settings to activate Windows.
07:22 PM
05-May-25



abc.java ➔ compilation ➔ abc.class ➔ run ➔ jvm ➔ interpreted

why c is a general / multipurpose programming language?

Using C language we can develop the software like

1. **Operating systems**

   Eg: windows, unix, linux, mac, ios, android,…

2. Editors

   Eg: notepad, wordpad, ms-word,….

3. Translators

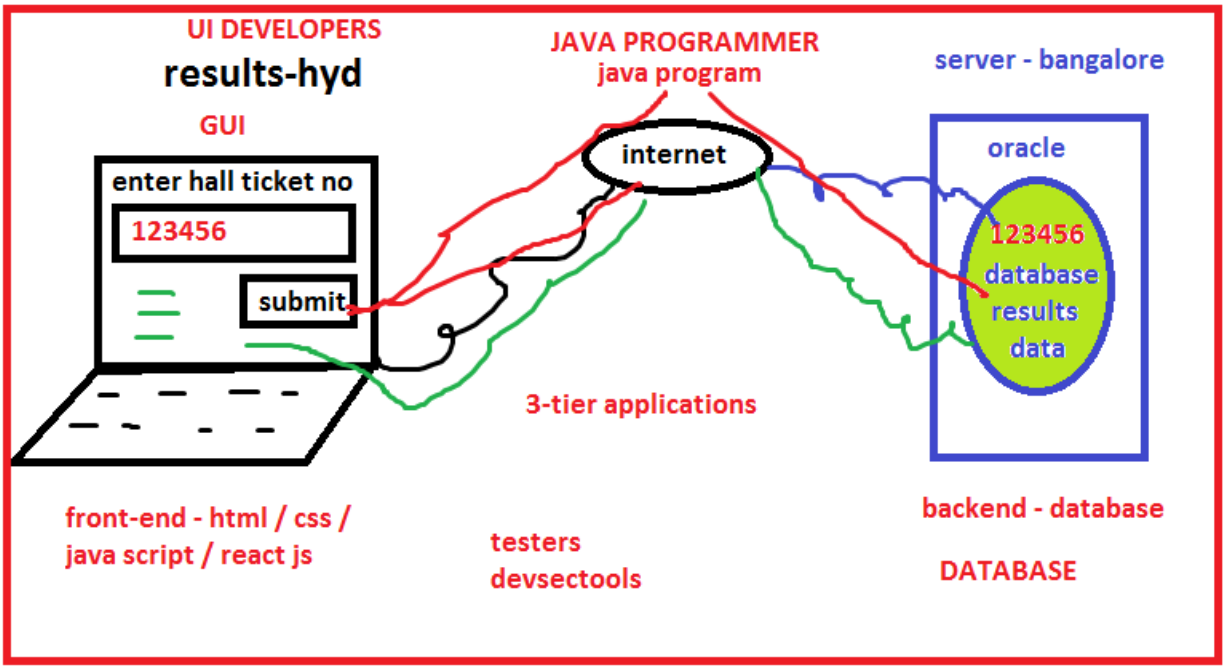
   Eg: compiler, interpreter, assembler

4. Commercial applications

   Eg: Hotel / super market / google pay/ atm,…

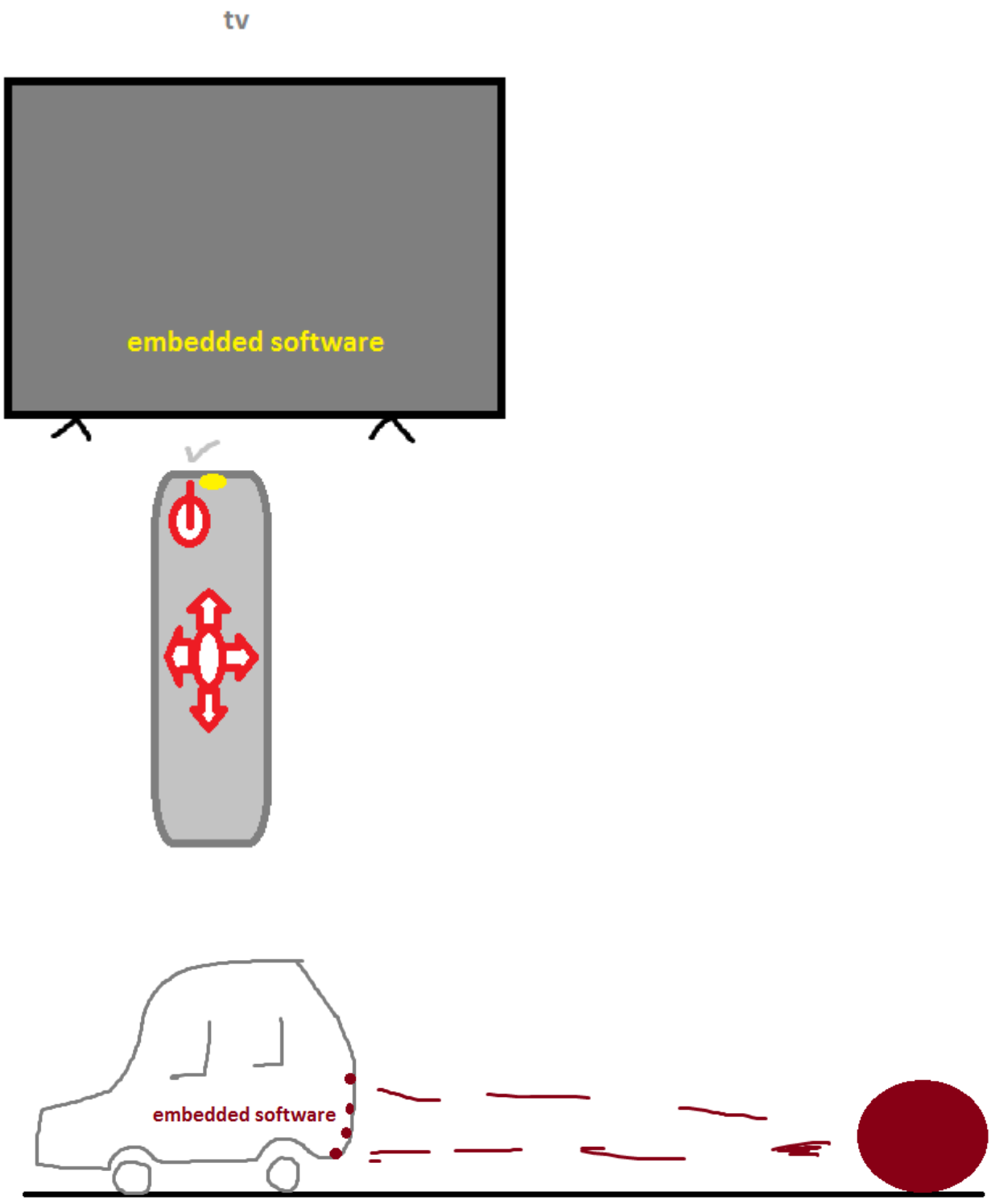
5. Data base

   Eg: oracle, sql server, my sql, dbII, mongodb,..



project work

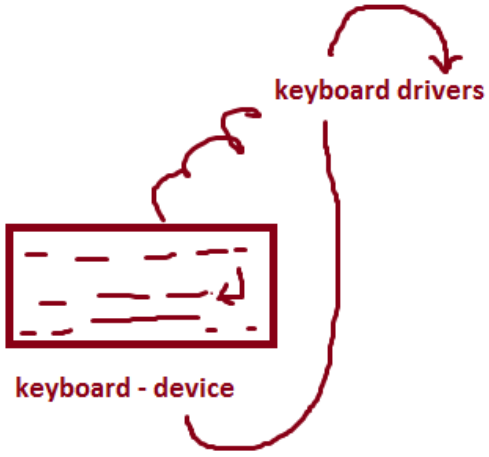
6. Embedded applications



7. **Device drivers**

   Eg: audio/ video / usb drivers / printer drivers,..





8. Pc & mobile games
9. Antivirus

   Eg: Norton, avast, macfee,….
10. Browsers

    Chrome, firefox, opera,….
11. Media players

    Eg: vlc, mx-player, windows media players,…
12. Standalone applications

What is programming paradigm?

Every programming language having a particular structure with rules and regulations. As a programmer

we should have to follow this and technically it is called programming paradigm.

Old languages are following monolithic programming

C is following procedure oriented programming

C++, python following procedure oriented and object oriented. Hence they are called multi-paradigm programming languages.

Java is object oriented programming language.

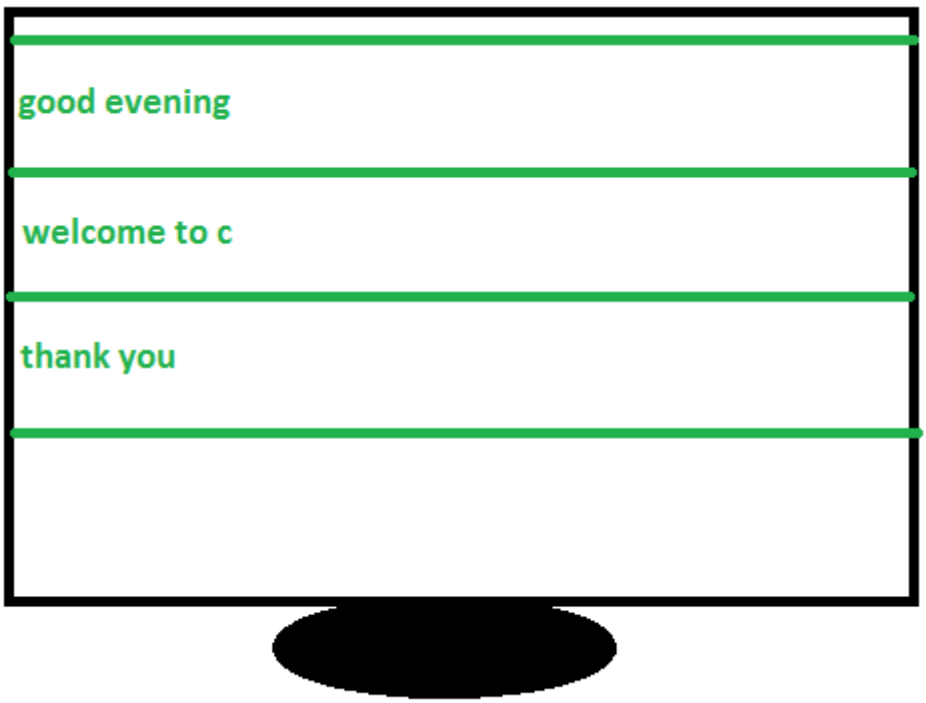
What is monolithic programming structure?

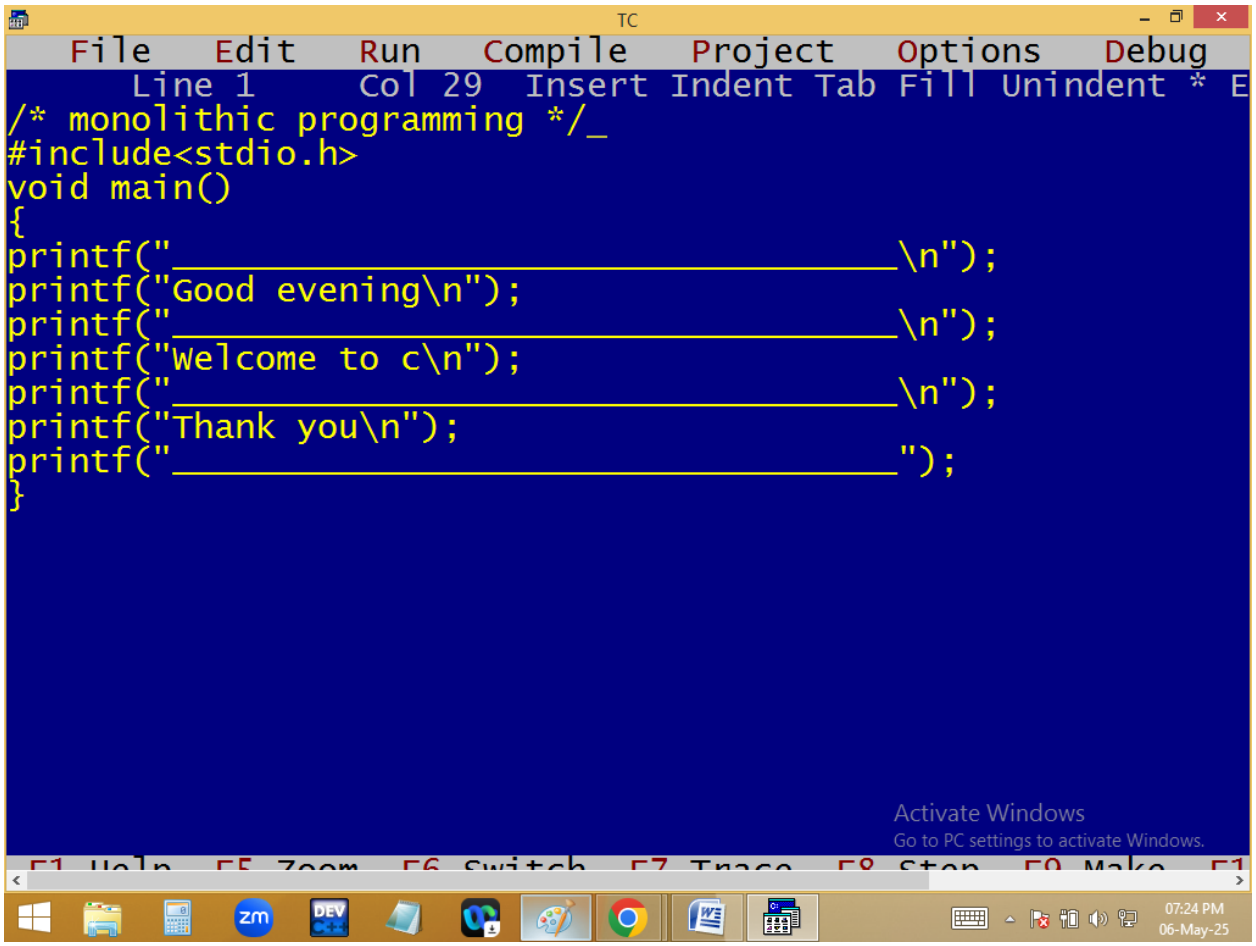
In this model the total program created with single program.

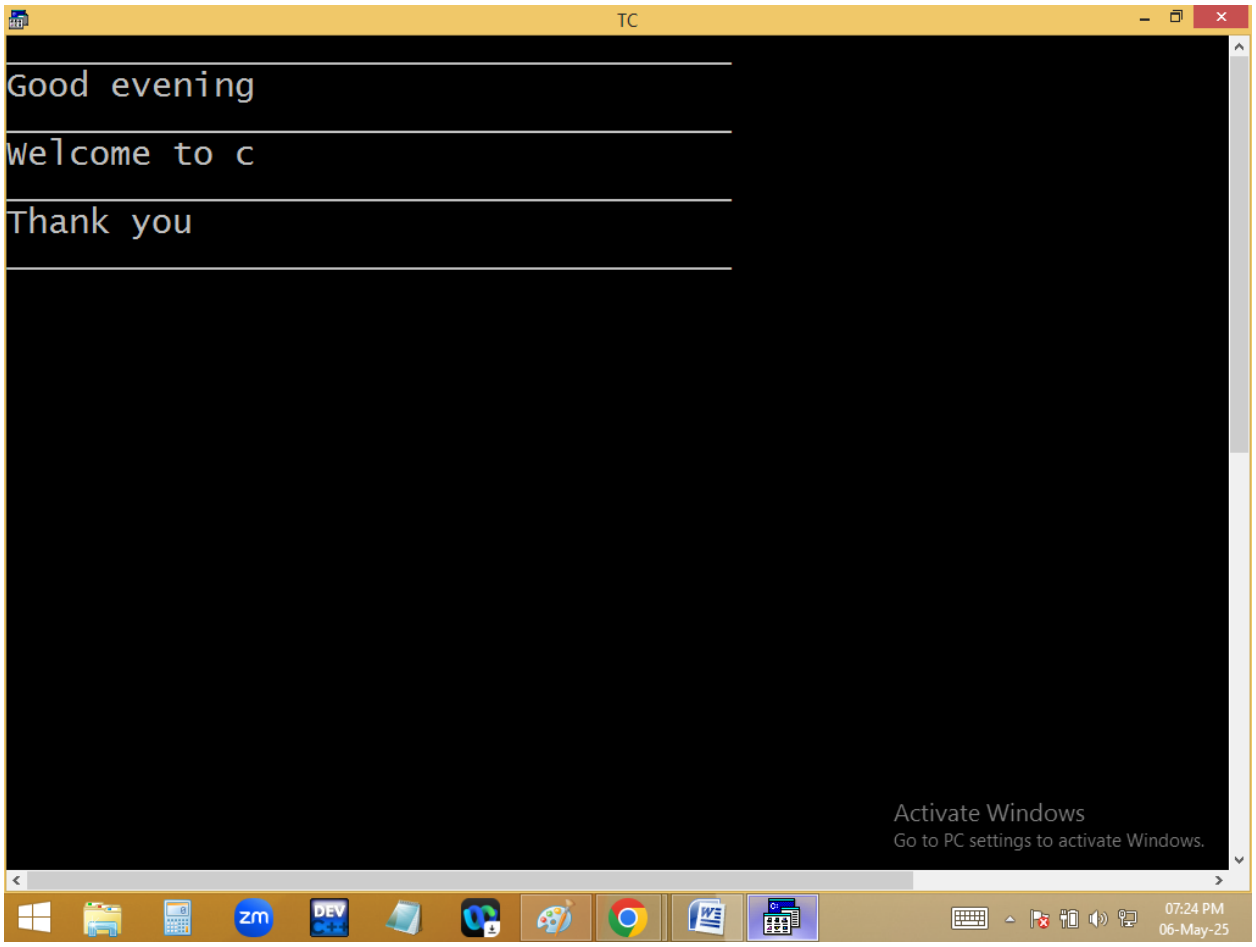
Disadvantages:

1. **Difficult to find the errors.**
2. **Program is big.**
3. **Takes more memory.**
4. **Performance is poor**
5. **No reusability.**

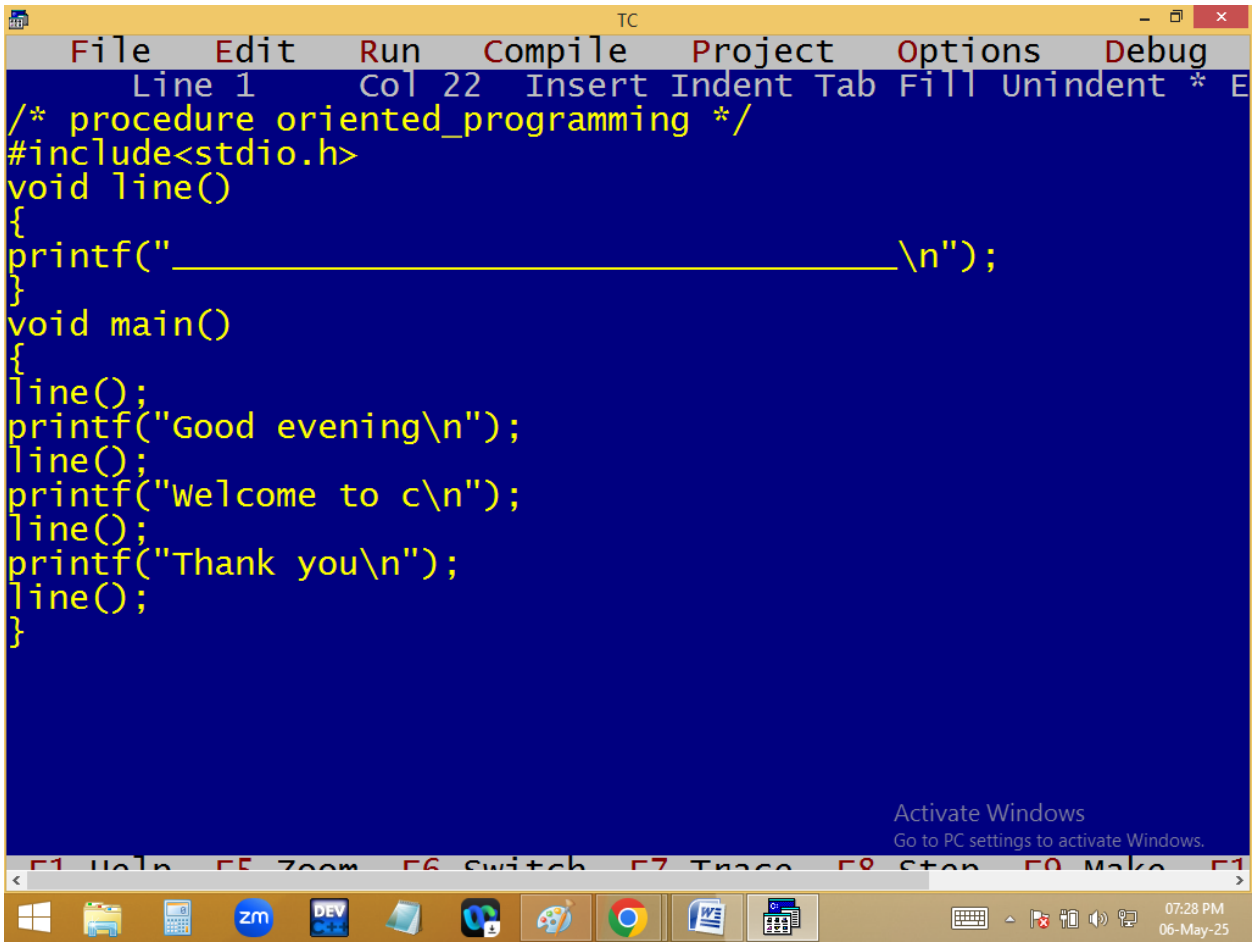
Example:

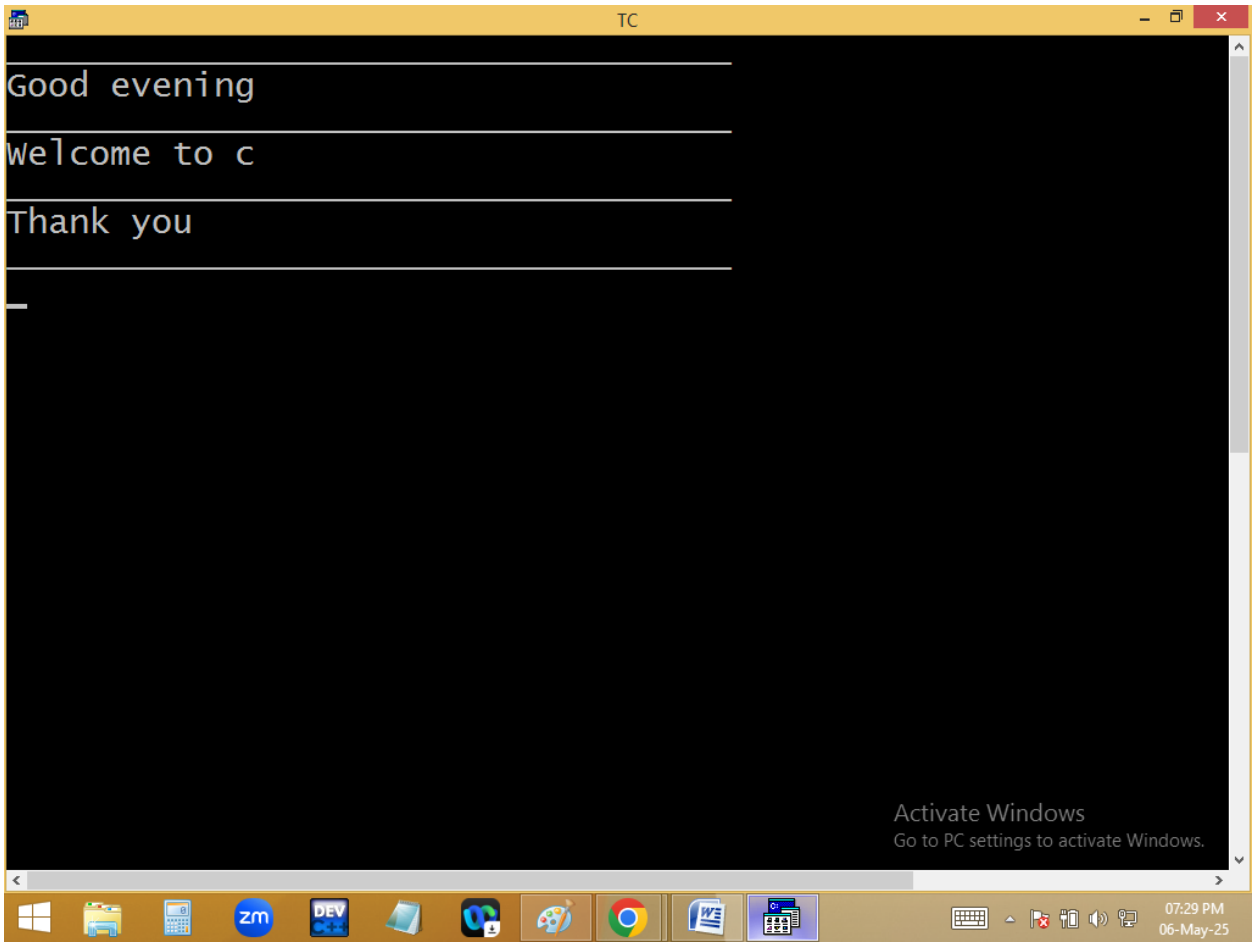

good evening
welcome to c
thank you

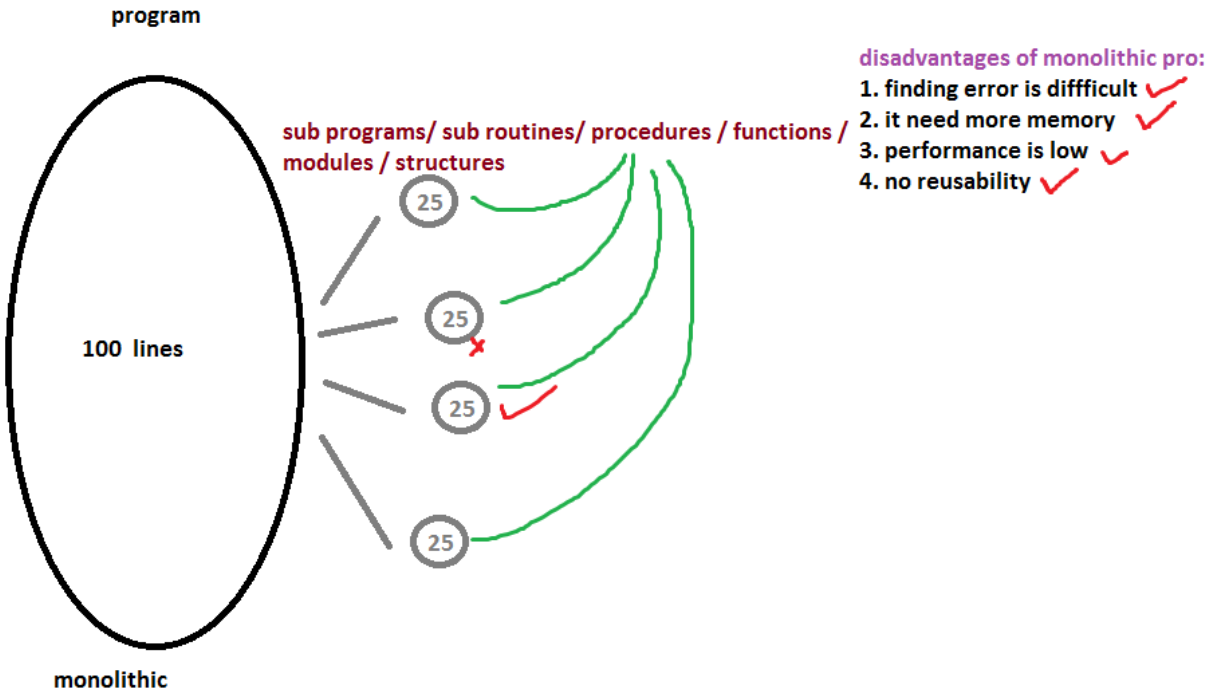

TC
File Edit Run Compile Project Options Debug
Line 1 Col 29 Insert Indent Tab Fill Unindent * E
/* monolithic programming */_
#include<stdio.h>
void main()
{
printf("_____\n");
printf("Good evening\n");
printf("_____\n");
printf("Welcome to c\n");
printf("_____\n");
printf("Thank you\n");
printf("_____");
}
Activate Windows
Go to PC settings to activate Windows.
07:24 PM
06-May-25



What is procedure oriented programming [POP]?


TC
File Edit Run Compile Project Options Debug
Line 1 Col 22 Insert Indent Tab Fill Unindent * E
/* procedure oriented_programming */
#include<stdio.h>
void line()
{
printf("_____\n");
}
void main()
{
line();
printf("Good evening\n");
line();
printf("Welcome to c\n");
line();
printf("Thank you\n");
line();
}
Activate Windows
Go to PC settings to activate Windows.
07:28 PM
06-May-25


TC
Good evening
Welcome to c
Thank you
Activate Windows
Go to PC settings to activate Windows.
07:29 PM
06-May-25


program
100 lines
monolithic
sub programs/ sub routines/ procedures / functions / modules / structures
25
25
25
25
disadvantages of monolithic pro:
1. finding error is diffficult
2. it need more memory
3. performance is low
4. no reusability

Pop advantages:

1. **Modularity: dividing big program into small pieces according to the program requirement.**
2. **Simplicity: easy to read and understand.**
3. **Reusability: write once, use many times.**
4. **Performance: efficiency improved.**